Problem Statement

Educational institutions need effective methods to identify students who may experience academic difficulties before final grades are released. Predicting student performance can help teachers and administrators provide timely academic support and interventions.

This project aims to develop a machine learning regression model capable of predicting a student's final grade (G3) using demographic, family, social, and academic factors from the UCI Student Performance Dataset developed by Cortez and Silva (2008). The study evaluates multiple regression algorithms and determines which model provides the most accurate predictions.

Accurate grade prediction can assist educational institutions in:

- Identifying students who may need additional academic support.
- Monitoring student progress throughout the academic year.
- Supporting data-driven educational decisions.
- Improving student retention and academic outcomes through early intervention.

Rather than waiting for final examination results, educators can use model predictions to recognize potential performance issues earlier and provide assistance when it is most needed.

Data Source

The dataset used in this study was obtained from the UCI Machine Learning Repository's Student Performance Dataset. The dataset was originally collected by Cortez and Silva (2008) to study factors influencing student achievement in secondary education.

The dataset contains academic, demographic, social, and family characteristics of secondary school students enrolled in Mathematics courses.

Dataset Size

- Total Records: 395 students
- Total Variables: 33 original attributes
- Target Variable: G3 (Final Grade)

Target Variable

The target variable is:

G3 = Final Grade

Grades range from:

0 to 20

A regression approach was selected because the target variable is a continuous numerical value rather than a category.

What Is Missing

Several potentially important educational factors are not available in the dataset, including:

- Teacher evaluations
- Classroom participation
- Learning styles
- Psychological factors
- Detailed socioeconomic indicators
- Attendance records beyond total absences

These variables may influence student performance and could further improve prediction accuracy if included in future studies.

Preprocessing Decisions and Reasons

Several preprocessing steps were applied before model training.

Removal of Missing Data Issues

A check for missing values was performed.

Result:

> No missing values were found in the dataset.

Since the dataset contained complete records, no imputation or data filling procedures were necessary.

Categorical Variable Encoding

Several attributes contained categorical values, including:

- School
- Sex
- Address
- Parent occupations
- Guardian

These variables were transformed using One-Hot Encoding.

Reason:

Machine learning algorithms require numerical input data. One-Hot Encoding converts categorical variables into machine-readable numeric representations while minimizing unintended ordinal relationships (Harrison, 2020).

Feature and Target Separation

The target variable, **G3**, was separated from predictor variables.

Reason:

The target variable represents the outcome being predicted and must not be included among the input features because doing so would cause target leakage and artificially inflate model performance.

Train-Test Split

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

Reason:

This allows the model to learn patterns from training data while being evaluated on previously unseen observations, providing a more realistic measure of predictive performance (Géron, 2022).

The Model Applied

Three regression algorithms were evaluated.

Linear Regression

Linear Regression is a statistical learning algorithm that assumes a linear relationship between predictor variables and the target variable.

Advantages:

- Simple implementation
- Easy interpretation
- Computationally efficient

(James et al., 2021)

Decision Tree Regressor

Decision Tree Regression predicts outcomes by recursively partitioning the dataset into smaller groups based on decision rules.

Advantages:

- Easy visualization
- Handles nonlinear relationships
- Requires minimal data preparation

(Géron, 2022)

Random Forest Regressor

Random Forest is an ensemble learning algorithm that combines multiple decision trees and aggregates their predictions.

Advantages:

- Reduces overfitting
- Handles complex nonlinear relationships
- Typically achieves higher predictive accuracy
- Performs well on datasets with mixed feature types

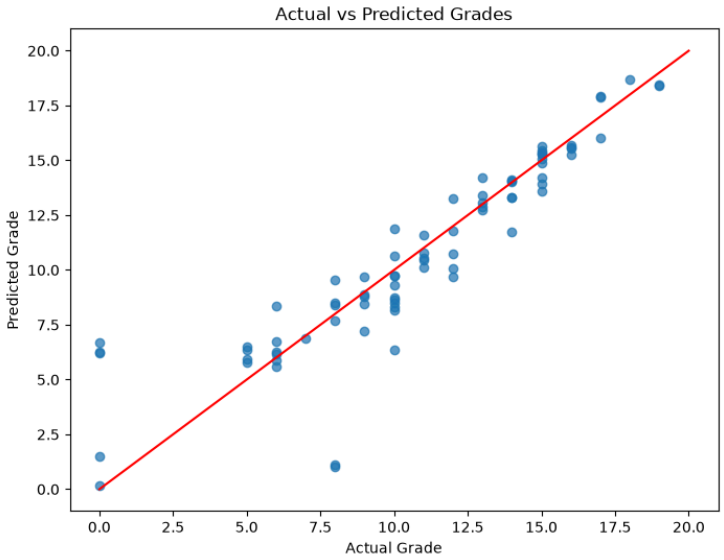
(Breiman, 2001)

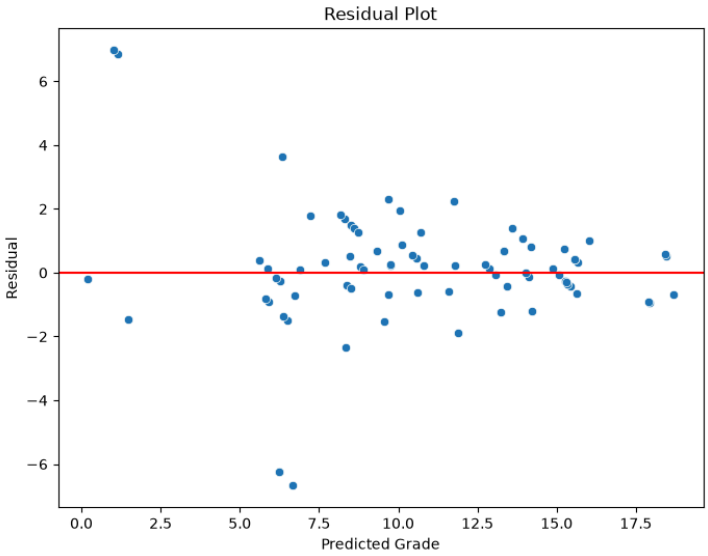
Final Model Selection

The Random Forest Regressor was selected as the final model because it achieved the highest coefficient of determination ($R^2 = 0.815$) and the lowest prediction error among the evaluated models. These results indicate superior predictive performance compared with Linear Regression and Decision Tree Regression.

Model Comparison

Model	MAE	RMSE	R^2
Linear Regression	1.647	2.378	0.724
Decision Tree	1.139	2.050	0.795
Random Forest	1.165	1.949	0.815





Actual vs. Predicted Grades

The Actual vs. Predicted Grades plot demonstrates a strong positive relationship between observed and predicted final grades. Most observations are located close to the diagonal reference line, indicating a high level of agreement between predicted and actual values. This suggests that the Random Forest Regressor is capable of producing accurate grade estimates for most students. According to Géron (2022), predictions that closely follow the ideal prediction line indicate strong model performance and good predictive accuracy.

Residual Plot

The residual plot shows that most residuals are distributed around the zero line, indicating that the model does not consistently overestimate or underestimate student grades. No strong systematic pattern is visible, which suggests that the model captures most of the important relationships within the dataset.

However, several larger residuals are observed among students with very low grades, indicating that prediction errors tend to be larger for low-performing students. In contrast, residuals for students with average and high grades are generally smaller and more evenly distributed. According to James et al. (2021), randomly distributed residuals around zero are desirable because they indicate that the model errors are largely random rather than systematic.

Interpretation: What the Numbers Mean for the User

The Random Forest Regressor achieved a Mean Absolute Error (MAE) of **1.165**, a Root Mean Squared Error (RMSE) of **1.949**, and an R^2 score of **0.815**.

The MAE indicates that the model's predicted grades differ from actual grades by approximately **1.17 grade points on average**. Since the target variable (G3) is measured on a scale ranging from 0 to 20, this error is relatively small and suggests good predictive performance.

The RMSE of **1.949** indicates that most predictions are within approximately two grade points of the actual grade. Because RMSE penalizes larger errors more heavily than MAE, it provides additional insight into the magnitude of prediction errors (Géron, 2022).

The R^2 score of **0.815** indicates that approximately **81.5% of the variation in students' final grades** can be explained by the predictor variables included in the model. This represents a strong predictive relationship between the input features and academic performance (James et al., 2021).

For educators and school administrators, this level of accuracy provides a useful estimate of student academic performance before final grades are officially released. The model may therefore serve as a decision-support tool for monitoring student progress and identifying learners who may benefit from academic intervention.

Limitations and Future Improvements

Limitations

Several limitations were identified in this study:

1. The dataset contains only 395 student records, which may limit the model's ability to generalize to larger populations.
2. The data originates from a particular educational setting and may not accurately represent students from different schools, regions, or educational systems.
3. Important factors such as motivation, classroom participation, learning styles, teacher assessments, and psychological characteristics are not included in the dataset.
4. Previous grades (G1 and G2) are among the strongest predictors of the final grade (G3). While this improves predictive accuracy, it may reduce the model's effectiveness for very early intervention before sufficient academic performance data become available.

Future Improvements

Given more time and access to additional data, future work could:

- Collect larger and more diverse educational datasets.
- Include attendance records, behavioral indicators, and classroom participation metrics.
- Incorporate psychological and motivational variables that may influence academic performance.

- Evaluate advanced machine learning algorithms such as Gradient Boosting, XGBoost, and LightGBM.
- Develop a web-based or dashboard-based decision support system that provides real-time predictions for educators and administrators.

Conclusion

This study successfully developed a machine learning regression model for predicting student final grades using demographic, family, social, and academic information from the Student Performance Dataset developed by Cortez and Silva (2008).

Among the evaluated algorithms, the Random Forest Regressor achieved the best overall performance, obtaining an R^2 score of **0.815**, an MAE of **1.165**, and an RMSE of **1.949**. These results indicate that the model can explain a substantial proportion of variation in student achievement while maintaining relatively small prediction errors.

The findings demonstrate that machine learning techniques can effectively estimate student academic performance and support data-driven educational decision-making. Although the model has limitations, it provides a practical foundation for future academic support systems and early intervention initiatives.

References

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